Zhalok Rahman

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Video Introduction: https://www.loom.com/share/c5ab46a4737244edafd9f052555767a1

Skills

Golang, Python (FastAPI, Asyncio, Pytorch, Pandas), Node JS (Nest JS, Express JS, Koa), JavaScript/TypeScript (React JS, Next JS, PostgreSQL, MongoDB, Redis (Cache, Pub/Sub, Streams), RabbitMQ, WebSocket, SSE, Docker, AWS (ECS, CloudWatch, SQS, Kinesis, Lambda, OpenSearch, DynamoDB), Grafana, Prometheus, CI/CD (Github Actions, CodePipeline

Education

Shahjalal University of Science and Technology
Bachelors of Science in Computer Science
2019-01-01 – 2024-04-01

Experience

Kinetik Healthcare *Software Engineer 2*
2025-04-01 – Present

- Built **Florence** an **AI voice assistant** which handles around 80 percent of our ride initiation requests
- Developed internal **VoIP** testing tools for simulating calls for the voice agent
- Built a source control pipeline to manage **11Labs agent as code**
- Managed **CI/CD pipeline** for deploying agent
- Integrated Intercom based human agent call center with AI call center for transferring calls as fallback
- Built **Grafana dashboard** for monitoring the AI agent performance
- Contributed to **Langgraph** based orchestrator service for a large scale **AI Agent**
- Enhanced realtime notification pipeline with push based mechanism with **Redis Pub/Sub**, **CDC**, **Redis Streams**
- Migrated legacy code from **Express JS** to **Nest JS** for cleaner domain driven design
- Enhanced existing **RBAC** system with realtime access revocation
- Investigated production incidents based on logs and traces from **CloudWatch** and **Grafana**
- Optimized MongoDB query performance by **41 percent** using covered atlas search index scan
- Reduced network content download time by **66 percent** with default query projection
- Conducted AI-driven development using **Kiro**

PropertyFinder *Software Engineer*
2024-10-01 – 2025-10-31

- Ensured reliability and performance of **Golang** based B2B platform
- **Traced race conditions** between distributed services that was causing p1 incidents and confusing user experiences
- Investigated and fixed root cause of critical production incidents using metrics, logs, and traces from **Grafana**
- Conducted **Load Testing** and built dashboards to observe the performance of the entire system under high stress
- Written scripts for data fix to mitigate incidents
- Contributed to the modernization of B2B service, making architectural decisions and strategies for efficient migration
- Written unit tests to ensure **80 percent coverage**
- Designed asynchronous event-driven pipelines using **SQS** and **Kinesis Streams**
- Configured alerts in **Grafana** with **Slack** integration for real-time notification on any potential incident
- Used **Windsurf** for studying legacy codebases efficiently

Learnyx AI *Fullstack Engineer*
2023-10-01 – 2025-02-01

- Architected and built a scalable REST API with **NestJS**
- Designed an AI generation pipeline using **LangChain** for structured prompt management and multi-step chaining
- Integrated **Langfuse** for LLM observability, enabling cost tracking and performance tracing across model calls

- Decoupled heavy workloads from the main request cycle using asynchronous **RabbitMQ** consumers, improving resource efficiency
- Enhanced UX by implementing real-time streaming of AI-generated responses
- Optimized load performance using a hybrid lazy-loading and preloading strategy
- Acted as a technical advisor in stakeholder discussions, bridging engineering constraints with business goals
- Mentored a team of 3 engineers on system architecture patterns and backend best practices
- Managed self-hosted infrastructure on **AWS EC2** using **Coolify**, supporting **500+ active users**
- Integrated **Stripe** to handle recurring billing, subscription lifecycle, and payment management

Innovext IT *Fullstack Developer*
2022-12-01 – 2023-10-01

- Built features for online healthcare platform with NextJS, Express JS, MongoDB
- Integrated Stripe and Paypal payment gateways
- Synced client app in realtime with webhook event
- Implemented safe transactional credit transfer with MongoDB sessions
- Built YOLO inference pipeline with ONNX runtime

Certifications

Fundamentals of Operating Systems
Udemy
https://www.udemy.com/certificate/UC-ff0d44a3-8917-40cd-aef9-a01e258aee49/
Learned about Process structure, Memory managment, CPU cycles, I/O management

Fundamentals of Network Engineering
Udemy
https://www.udemy.com/certificate/UC-0a49d23e-036a-4c80-ad5a-410ccb877ebb/
Learned about under the hood mechanisms of network protocols

NodeJS Internals and Architecture
Udemy
https://www.udemy.com/certificate/UC-2068cbe0-b17c-47ca-b298-d8acb94b30e2/
Learned about internal architecture of node js, event loop, node js worker threads, enhancing node js app performance

Publications

Optical Mark Recognition with Object Detection and Clustering
2024 6th International Conference on Electrical Engineering and Information and Communication Technology (ICEEICT) *May 2024*
https://doi.org/10.1109/ICEEICT62016.2024.10534561

- Written scripts for generating synthetic training data
- Built training scripts with ultralytics for training lightweight YOLO models
- Implemented clustering based unsupervised algorithm for column detection and serialization of answers
- Exported finetuned model in Onnx format for inferencing via API
- demo: https://www.youtube.com/watch?v=BiEybE5qQqw

Detecting Communal Violence in Texts
Association for Computational Linguistics *May 2024*
https://aclanthology.org/2023.banglalp-1.25/

- Curated and annotated a large-scale dataset of violence-related texts aggregated from multiple heterogeneous sources
- Benchmarked multiple pretrained BERT variants to identify the optimal backbone for communal violence detection
- Applied domain-adaptive pretraining and fine-tuning to boost classification accuracy on low-resource Bangla text